Tumor-Associated Brain Frailty and Perioperative Brain Injury in Neuro-Oncologic Surgery: A Scoping Review

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Abstract

No instrument describes the vulnerable brain of a patient with a brain tumor. This scoping review mapped the literature to propose a working concept of tumor-associated brain frailty, chart its candidate domains, and ask whether neuroinflammation and perioperative brain injury are shared mechanisms in malignant and benign tumors. We followed Joanna Briggs Institute guidance and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews and searched six sources to March 2026. Evidence was mapped and tallied by direction of effect without pooling. From 2,476 records, 76 studies were included, 70 in a core evidence base and 6 in an instrument-focused stratum on the Indonesian Montreal Cognitive Assessment and Neurologic Assessment in Neuro-Oncology scale. No core study defined brain frailty. Frailty indices appeared in 7 core studies, cognitive testing in 30, neurological outcomes in 16, and imaging in 32. Cognition was impaired before surgery in all 8 studies that tested it, and new diffusion-weighted lesions followed 19% to 46% of resections. S100B, glial fibrillary acidic protein, and neurofilament light chain rose after surgery, but no core study reported meningioma-specific cytokine data. None used the Indonesian instruments. Tumor-associated brain frailty is best framed as a four-domain construct (systemic, cognitive, neurological, and imaging) with a two-hit course, in which the tumor is the first hit and perioperative brain injury the second. The evidence is observational, and the interaction between the hits is untested. Prospective cohorts that pair preoperative profiling with postoperative injury markers are the next step.

Key words: Biomarkers, Brain Neoplasms, Frailty, Neuroinflammatory Diseases, Postoperative Cognitive Complications

Introduction

Brain tumors are a common cause of neurological and cognitive morbidity,[1,2] and many patients with them undergo craniotomy. Cognitive deficits are present at diagnosis, before any treatment has been given.[3] The operation then adds anesthetic exposure, retraction, vascular manipulation, and a surgical inflammatory response to a brain whose reserve is unknown to the team in the operating room.

Frailty is the concept clinicians have borrowed to describe that reserve. In geriatrics it denotes a loss of physiological reserve that increases vulnerability to stressors.[4-6] Neurosurgical oncology has adopted general frailty indices, and a meta-analysis of the modified frailty index linked higher scores to worse perioperative outcomes in intracranial tumor surgery.[7] Two systematic reviews, however, stress what the field still lacks. One found that studies of frailty in brain tumor surgery used different instruments and thresholds and that best practice for assessing frailty in these patients remains unclear.[8] The other reported that the literature groups all tumor types together and so overlooks differences between etiologies.[9] Neither review identified a nomenclature or an index built for the brain of a patient with a tumor.

Neurology complicates the picture because frailty is used for several different things. Imaging-defined brain frailty, built from leukoaraiosis, atrophy, and old vascular lesions, predicts outcome after acute stroke.[10] Cognitive frailty couples physical frailty with cognitive impairment short of dementia.[11] A patient with a glioma or a meningioma may meet neither definition and still carry a vulnerable brain, because the tumor adds mass effect, edema, barrier disruption, and local inflammation. We use the working term tumor-associated brain frailty for this state and treat it as a concept to be mapped.

Neuroinflammation is the most plausible shared mechanism. Microglia and macrophages are central components of the glioma microenvironment,[12,13] but meningiomas, which are usually benign, also carry immune infiltrates that correlate with tumor behavior and peritumoral edema.[14,15] Surgery and anesthesia then trigger a second inflammatory wave that is increasingly linked to perioperative neurocognitive disorders.[16,17] Microglial priming studies show that a brain already exposed to inflammation can react more strongly to a later systemic challenge,[18,19] which suggests a two-hit sequence in which the tumor is the first hit and perioperative brain injury the second. Circulating markers such as S100B, glial fibrillary acidic protein (GFAP), and neurofilament light chain (NfL) could make the second hit measurable.[20,21]

Two instruments bear directly on the cognitive and neurological domains. The Montreal Cognitive Assessment (MoCA) detects cognitive impairment that the Mini-Mental State Examination (MMSE) misses in patients with brain metastases,[22,23] and the Neurologic Assessment in Neuro-Oncology (NANO) scale was designed to give clinicians a structured measure of neurologic function.[24] Indonesian versions of both exist, but their use in perioperative neuro-oncology is unclear.

We therefore mapped the literature on patients with intracranial tumors undergoing surgery with seven review questions (RQ). How is frailty or brain frailty defined and measured in this population (RQ1)? Which domains and instruments characterize it (RQ2)? Is neuroinflammation documented in benign as well as malignant tumors (RQ3)? Which biomarkers can detect neuroinflammation and brain injury (RQ4)? What does the evidence say about perioperative brain injury as a second hit (RQ5)? Which cognitive and neurological instruments have been used, and what exists for the Indonesian MoCA (MoCA-INA) and NANO (NANO-Ina) (RQ6)? Where are the gaps (RQ7)?

Methods

Review Design and Reporting

We conducted a scoping review to map the breadth and characteristics of evidence on tumor-associated brain frailty, its candidate domains, and the mechanisms and markers of perioperative brain injury in patients with brain tumors. We followed the Joanna Briggs Institute (JBI) methodology for scoping reviews,[25] which builds on the framework of Arksey and O'Malley and its refinement by Levac et al.[26,27] Reporting follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension for scoping reviews (PRISMA-ScR) and the PRISMA 2020 flow diagram format[28,29] (Supplemental Digital Content 1, Table showing the completed PRISMA-ScR checklist). A scoping approach was chosen because the aim was to map concepts and gaps and not to estimate pooled effects.[30] The review was not registered in a prospective registry. Ethics approval was not required because the review used published data only. The search strategies, charting form, coded dataset, and analysis scripts are deposited in a public GitHub repository (https://github.com/sharing-123/tumor-associated-brain-frailty-and-perioperative-brain-injury-scoping-review), and an archived version is deposited on Zenodo (https://doi.org/10.5281/zenodo.23443881).

The seven review questions were set with the clinical team commissioning the review. At the consultation stage,[27] the team asked for evidence on the Indonesian versions of the MoCA and the NANO scale, which had not been part of the original search. This produced a post hoc amendment, the instrument-focused stratum described below. A question could end partly answered or unanswered, which in a scoping review is itself a finding.

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Eligibility Criteria

Eligibility followed the Population, Concept, and Context framework.[25] The population comprised patients of any age with intracranial tumors, primary or metastatic and benign or malignant, who were scheduled for or had undergone surgery or whose tumor tissue was obtained at surgery. The concept covered frailty or vulnerability of the brain and body, assessed through frailty indices, cognitive testing, neurological assessment, and imaging, together with neuroinflammation, brain injury biomarkers, perioperative brain injury, delirium, and anesthetic strategies evaluated against these outcomes. The context was perioperative neurosurgical neuro-oncology in any setting and country.

Eligible designs were trials, trial protocols, cohorts, registry analyses, case-control and cross-sectional studies, case series, and human mechanistic studies. We excluded studies on traumatic brain injury, stroke, or non-tumor conditions, studies of chemotherapy or radiotherapy without surgery or anesthesia, studies that reported no outcome related to cognition, neuroinflammation, or injury biomarkers, and case reports, letters, commentaries, narrative reviews, conference abstracts, and reports without an English full text. The instrument-focused stratum admitted primary studies of MoCA-INA, NANO-Ina, or their parent instruments in neuro-oncology populations, and studies of MoCA-INA in any adult population. One Indonesian-language report with an English abstract was admitted. This stratum was reported separately and did not contribute to the tumor-specific counts.

Search Strategy

We searched PubMed, Scopus, ProQuest, and the Cochrane Library (CENTRAL) from inception, with medRxiv and Google Scholar as additional sources. The last searches were run on March 22, 2026. Each strategy combined three blocks, covering tumor terms (for example, glioma and meningioma), surgery and anesthesia terms (for example, craniotomy and total intravenous anesthesia), and terms for frailty, perioperative brain injury, neuroinflammation, the blood-brain barrier, injury biomarkers such as S100B, neuron-specific enolase (NSE), and GFAP, postoperative cognitive dysfunction, delirium, and neuroprotection. The database-specific strategies are given (Supplemental Digital Content 2, Table showing the database-specific search strategies and record counts). The search strategy was not peer reviewed. Citation searching of included studies added further records, and the clinical team supplied seven sources on MoCA-INA and NANO-Ina.

Study Selection and Data Charting

Duplicates were removed before screening. Two reviewers screened titles and abstracts independently against the eligibility framework, and the same two reviewers assessed full texts with a recorded reason for each exclusion. Every disagreement at either stage was resolved by discussion in a video meeting. During data charting, ten further studies were found not to meet the framework. They were moved to the full-text exclusion counts with a reason, so the final numbers differ from the original screening tally of 80 included studies (Supplemental Digital Content 3, Table reconciling the original screening tally with the final included studies).

A structured charting form with 31 items was applied to every included study (Supplemental Digital Content 4, Table showing the 31-item charting form for all 76 studies). Items covered bibliographic data, design, tumor type, sample size, imaging, biomarkers with

source and timing, hemodynamic measures, frailty indices, neuropsychological tests, and operational definitions of outcomes, together with key findings. Key findings were restricted to results that carried a reported statistic so that each entry could be checked against the source. An item not evaluated in a study was recorded as not assessed, which keeps absent data distinct from an extraction gap.

Evidence Classification and Synthesis

Each study was assigned a design class and an evidence level from the Oxford Centre for Evidence-Based Medicine 2011 framework, applied to design alone. Randomized trials and prospective cohorts were graded level 2, retrospective cohorts, registry analyses, and secondary analyses of trials level 3, case-control studies, cross-sectional studies, case series, and non-randomized pilot trials level 4, and mechanistic laboratory studies level 5. Trial protocols were left ungraded. The level therefore describes design and does not judge quality. Each study was also placed in one of six evidence streams and coded to four candidate frailty domains (systemic frailty, cognition, neurological function, and imaging) and to mechanistic domains (inflammatory markers, brain injury biomarkers, delirium, hemodynamics, and tumor biology) (Supplemental Digital Content 5, Table showing the coding of studies by evidence stream, design, tumor category, and domain). A study could contribute to more than one domain and was counted once within each.

Synthesis combined four methods. We tabulated study characteristics as a descriptive numerical summary, built an evidence map that places domains against tumor categories in the manner of an evidence and gap map,[31] and used framework-based content analysis, with the four domains as the a priori framework, to describe instruments and definitions. For questions about direction, we tallied the direction of effect that the authors reported at the group level, a form of

vote counting by direction of effect described in the Cochrane Handbook and the Synthesis Without Meta-analysis (SWiM) guideline.[32,33] These tallies describe agreement in direction and do not estimate an effect. Pooling was not attempted because outcomes, tumor types, and time points differed too widely. Formal risk-of-bias appraisal was not performed because critical appraisal is optional in scoping reviews,[25,28] and the design-based evidence level was used instead. Counts were generated from the coded dataset with Python 3.12.10, pandas 3.0.5, and matplotlib 3.11.1.

Results

Study Selection

The searches returned 2,318 records from four databases and medRxiv and 117 from Google Scholar. Citation searching added 34 records and the clinical team supplied 7 sources, so 2,476 records were identified. After 987 duplicates were removed, 1,331 records were screened by title and abstract and 1,062 were excluded. Of 269 reports sought, 252 were assessed in full text, 215 were excluded, and 37 were included. From the other sources, 145 of 158 reports were assessed, 106 were excluded, and 39 were included (Figure 1). Reasons for exclusion are given in the figure.

In total, 76 studies were included, comprising a core evidence base of 70 studies and an instrument-focused stratum of 6 studies. Ten studies from the original tally of 80 were removed during data charting, one for population (a spinal surgery trial) and nine for outcome.

Characteristics of the Core Evidence Base

The 70 core studies were published between 1987 and 2026, and 43 appeared from 2018 onward. They came from 21 countries, most often the United States (13) and China (13). One study came from Southeast Asia (Thailand) and none came from Indonesia. Designs were

prospective cohorts (28), randomized trials (17), retrospective cohorts (11), registry cohorts (3), and 11 other designs. By design, 45 studies (64%) reached level 2, 15 level 3, 6 level 4, and 2 level 5, and two protocols were ungraded. Study samples were glioma only in 25 studies, meningioma only in 10, brain metastasis only in one, and mixed tumor types in 34. Among the 68 studies that reported a sample size, the median was 60 (interquartile range, 32 to 164, range, 7 to 28,340). Six evidence streams emerged, namely tumor biology and the blood-brain barrier (8 studies), systemic frailty and perioperative risk (5), cognitive trajectories after tumor surgery (22), postoperative delirium or agitation (6), perioperative brain injury (10), and perioperative interventions or trial protocols (19). Study-level details are given (Supplemental Digital Content 6, Table and figure showing the characteristics of the 70 core studies).

How Frailty Is Defined and Measured

No core study used the terms brain frailty, cognitive frailty, or tumor-associated frailty in its full text. Frailty was operationalized with general instruments in seven studies. The five-factor modified frailty index (mFI-5) was used in three,[34-36] and the eleven-item modified frailty index,[37] the Hopkins Frailty Score,[38] the Risk Analysis Index,[36] the Fried phenotype,[39] and a comorbidity count used as a proxy[40] were each used once. Six of the seven studies sampled mixed tumor types and one sampled brain metastases, so no frailty study was restricted to glioma or meningioma. Cutoffs differed even for the same index.[34,35]

In all seven studies, higher frailty or comorbidity burden was linked to worse outcomes, or a frailty-directed intervention improved them. Postoperative delirium rose from 12.8% in non-frail to 27.0% in pre-frail and 37.5% in frail patients.[37] In a registry of 20,333 craniotomies, low and medium-to-high mFI-5 raised the odds of 30-day mortality (odds ratios, 1.87 and 2.42).[34] Frail patients had higher odds of non-home discharge,[38] and each mFI-5 point added 1.38

hospital days.[35] The Risk Analysis Index separated 30-day mortality of 1.3% in robust and 12.6% in severely frail patients with brain metastases.[36] A frailty-originated rehabilitation program lowered delirium from 28.7% to 15.6%,[39] and delirium was more frequent with three or more comorbidities.[40] Brain frailty in tumor patients is therefore defined today by borrowing general indices, and the mapped literature offers no brain-specific nomenclature or index.

Candidate Domains and Instruments

Table 1 summarizes the four domains, and the same domains appear with their instruments and gaps in a figure (Supplemental Digital Content 7, Figure showing the four candidate domains, instruments, and gaps). In the systemic domain, the seven studies above supplied the only instruments. In the cognitive domain, 30 core studies reported cognitive testing. Fourteen used multidomain batteries, four used domain-specific tests, four used computerized batteries, and seven used brief screeners. The MMSE appeared in five studies and the MoCA in two studies with data.[41,42] One study found that the Japanese MMSE stayed normal while detailed testing and lesion mapping exposed specific deficits.[43]

Cognition was impaired before surgery in all eight studies that compared patients with norms, controls, or cutoffs. At least one impaired test was found in 73.7% of patients with meningioma in one cohort[44] and 69% in another.[45] Across the 18 pre-post cohort studies, cognition improved in 6, did not change or showed a persistent deficit in 3, declined in 3, declined transiently in 3, and followed other patterns in 3. Of the 13 glioma-only cohorts, 8 reported an early or lasting decline and 2 reported improvement, whereas three of the four meningioma-only cohorts reported improvement. Among trials, postoperative MoCA did not differ with magnesium in an underpowered study,[41] whereas MoCA and the Cognitive Capacity Screening Examination were higher with dexmedetomidine on day 1.[42]

Sixteen core studies assessed neurological or functional outcomes. Eight recorded a binary new deficit, and the others used scales such as the Barthel index and the modified Rankin Scale. This makes nine instruments or definitions, none designed for neuro-oncology, and no core study used the NANO scale. Reported deficit rates differed with definition, from permanent deficits in 6.7% of meningioma operations[46] and 14% permanent motor deficits in glioma surgery[47] to early deterioration in 50% of meningioma patients.[48]

Imaging appeared in 32 core studies. Six used diffusion-weighted imaging (DWI) to detect new ischemic lesions, 13 used volumetric magnetic resonance imaging (MRI), four used perfusion or permeability imaging, seven used functional connectivity, and seven reported computed tomography (CT) findings. Imaging mostly described the tumor or documented injury. No core study derived an imaging frailty score from leukoaraiosis, atrophy, or old vascular lesions, no study measured temporal muscle thickness, and CT served to detect hematoma, edema, pneumocephalus, or midline shift and never to grade frailty.

Neuroinflammation in Malignant and Benign Tumors

Direct inflammatory evidence came from glioma and mixed samples (Supplemental Digital Content 8, Table showing inflammation, injury, and vulnerability evidence by tumor category). In high-grade glioma, CD68-positive macrophages (19% to 27% of cells) outnumbered CD45-positive leukocytes (3% to 7%), and cerebrospinal fluid interleukin-7 was lower than in controls.[49] In astrocytoma, interleukin-8 and interleukin-6 expression rose with grade and necrosis, and interleukin-8 independently predicted survival.[50] Peritumoral microdialysis showed that most cytokines peaked within 8 hours of surgery and fell over 96 hours.[51]

The ten meningioma-only studies measured no inflammatory marker. They contributed evidence of barrier-related vulnerability and injury instead. S100B rose after meningioma surgery in two cohorts and tracked bleeding, midline shift, and deterioration,[48,52] and 34.9% of meningioma operations produced a new DWI lesion.[46] No core study compared inflammatory markers between meningioma and glioma. The mapped studies support a shared vulnerability phenotype without testing a shared inflammatory mechanism.

Delirium risk also varied with histology. Five of the six studies that tested tumor type reported higher delirium risk with glioma or malignant tumors, and in one cohort meningioma was protective.[53] Two trials reported that dexmedetomidine and lidocaine lowered interleukin-6 and tumor necrosis factor alpha at the end of surgery or on day 1,[42,54] and in the dexmedetomidine trial the difference had converged with control by day 3.[42]

Biomarkers of Neuroinflammation and Brain Injury

The markers and their sampling are summarized (Supplemental Digital Content 9, Table showing the biomarkers measured in the core evidence base). S100B was the most frequent, with data in 10 studies, followed by NSE (7), GFAP (4), interleukin-6 (5), and vascular endothelial growth factor (3). Four observational studies followed injury markers after surgery and all four found a postoperative rise. Plasma GFAP, tau, and NfL rose on days 1 and 3, NfL showed a second peak at day 10, and day 1 GFAP and NfL were higher in patients with new deficits.[55] S100B above 0.4 μg/L carried a ninefold risk of neurologic deterioration.[48] GFAP rose after glioma surgery but did not track residual tumor.[56]

Eight randomized trials used S100B, NSE, or GFAP as endpoints. The intervention lowered at least one marker in six trials (electroacupuncture, magnesium in two, goal-directed fluid therapy, dexmedetomidine, and lidocaine). The primary S100B endpoint did not differ with

magnesium in one trial,[57] and S100B and NSE were higher with ciprofol than propofol in another.[58] Sampling was almost always peripheral, from baseline to 24 or 48 hours, and no perioperative trial measured a cytokine in cerebrospinal fluid. Injury markers therefore indicate that injury occurred, whereas the cytokine evidence needed to show neuroinflammation as such is compartment-specific and sparse.

Perioperative Brain Injury as a Second Hit

New DWI lesions followed resection in 34.9% of meningioma operations,[46] 44.2% of glioma operations,[59] 45.7% of glioma operations with motor monitoring,[47] and 18.8% of a mixed sample.[60] Among patients with a permanent deficit after meningioma surgery, 72.7% had an infarction,[46] and a new lesion predicted incomplete recovery at day 7 and day 30.[60] The other second-hit evidence is summarized (Supplemental Digital Content 10, Table showing the evidence on perioperative brain injury as a second hit).

Delirium and agitation occurred in 4.2% to 35% of patients in observational cohorts and in 28.7% to 46% of control arms, depending on instrument and definition. Hemodynamic findings were mixed. Intraoperative mean arterial pressure below 65 mmHg was not associated with delirium,[61] whereas a lower stroke volume variation threshold was linked to fewer new neurologic events (17.5% vs 40%).[62] Dexmedetomidine lowered delirium from 46% to 22%.[63]

Preoperative state predicted postoperative outcome in several designs. Presurgical cognition was the strongest predictor of early postsurgical cognition,[64] frailty strata predicted delirium,[37] and presurgical connectivity predicted later working memory.[65] Four trials tested magnesium as a protective agent. Two found lower S100B,[66,67] one found no primary difference,[57] and one found no difference in MoCA.[41] No core study examined the interaction

between a preoperative vulnerability measure and a postoperative injury marker in the same patients.

Instrument-Focused Stratum

None of the 70 core studies used MoCA-INA or NANO-Ina, and none evaluated these instruments before and after surgery. The instrument-focused stratum added six studies (Supplemental Digital Content 11, Table showing the instrument-focused stratum). The NANO scale covers nine neurologic domains, showed more than 90% inter-observer agreement (kappa, 0.35 to 0.83), and took a median of 4 minutes to administer.[24] Its Indonesian version was tested in 30 patients with brain tumors and reached kappa values from 0.629 for sensation to 0.935 for facial strength.[68]

For cognition, a feasibility study in 71 patients with primary central nervous system tumors found a median MoCA of 27, with 32% abnormal at a cutoff of 25 or lower, and lower scores accompanied lower Karnofsky status, older age, and greater self-reported cognitive symptom burden.[69] In 33 Indonesian patients with brain tumors assessed at diagnosis, mean scores were 21.79 on the MoCA-INA and 23.76 on the MMSE.[70] In 83 older adults without a tumor, MoCA-INA scores averaged 21.06 vs 24.96 for the MMSE (correlation, 0.71).[71] In 31 anesthesiology residents, mean MoCA-INA fell from 26.84 to 25.00 after a 24-hour on-call period.[72] Together these studies show that both instruments can be administered in Indonesian clinical settings, but none tested responsiveness to craniotomy.

Integrated Evidence Map and Conceptual Synthesis

Figure 2 places each domain against tumor category and shades each cell by the share of graded studies at level 2. Cognitive testing and imaging carried the most evidence and were concentrated in glioma and meningioma cohorts. Systemic frailty indices, delirium, and

hemodynamic studies came almost entirely from mixed samples. Inflammatory markers and tumor biology came from glioma and mixed samples, and none came from meningioma-only samples.

The synthesis supports a four-domain construct with a two-hit course (Figure 3). The tumor produces mass effect, edema, barrier disruption, and local inflammation, visible in the mapped evidence as impaired cognition, frailty-linked risk, and preoperative neurological and imaging abnormalities. Surgery and anesthesia then add ischemic injury, marker release, and delirium. The mapped studies describe each hit and show that preoperative state predicts postoperative course, but none tested the interaction.

Discussion

This review mapped 76 studies to ask what is known about the vulnerable brain of a patient with a brain tumor who faces surgery. The main finding is an absence. No study defined tumor-associated brain frailty, and the seven studies that used frailty at all borrowed indices built for older adults or general surgical patients. Around that absence sits a body of work on the four domains we propose as its components, on candidate mechanisms, and on perioperative injury, which can be assembled into a working construct but not yet a validated one (Table 2).

We propose tumor-associated brain frailty as a reduced capacity of the brain and body of a patient with an intracranial tumor to tolerate surgical and anesthetic stress. It is expressed in systemic, cognitive, neurological, and imaging domains and is driven at least in part by tumor-related inflammation and barrier injury. The definition differs from imaging-defined brain frailty and from cognitive frailty in three ways.[10,11] It is tied to a lesion and its mechanisms, so mass effect, edema, and barrier leak belong to it. It may be partly reversible, because improvement after resection was the commonest conclusion in meningioma cohorts and appeared in some

glioma cohorts. And it is defined against a specific coming challenge, the operation. These features justify a separate term, but the term stays a hypothesis until its components are shown to predict outcome beyond age and tumor grade, which none of the mapped studies did.

Each domain has instruments and no shared standard. The systemic indices reflect comorbidity and function more than brain reserve, and none of the frailty cohorts was restricted to glioma or meningioma, so tumor-specific thresholds cannot be derived from them. This extends the conclusions of earlier systematic reviews.[8,9] Cognition is the best-studied domain and also the least standardized. Impairment before surgery was universal among studies that looked for it, but declines clustered in glioma cohorts and often resolved, whereas meningioma cohorts mostly improved, which suggests that part of the preoperative deficit is a reversible tumor effect. The choice of instrument matters, because one study found normal screening scores in patients whose detailed testing showed specific deficits.[43] Neurological outcome was recorded with nine different instruments or definitions, and the NANO scale appeared only in the instrument-focused stratum.[24] Imaging was abundant but never converted into a frailty metric. Leukoaraiosis and atrophy, which are established in stroke,[10] were absent from the mapped set, as was temporal muscle thickness, a muscle-based measure linked to survival in glioblastoma.[73] Both are candidates for a CT or MRI component.

Whether neuroinflammation reaches beyond malignant tumors can be answered only in part. External reviews describe macrophage-rich immune infiltrates in meningioma that track edema,[14,15] and the mapped meningioma studies show injury and vulnerability in the form of S100B release, new DWI lesions, and cognitive deficits before surgery. None of them measured a cytokine. The cytokine data that exist relate to tumor grade and early postoperative kinetics and not to frailty. The similar vulnerability phenotype in benign and malignant tumors is therefore a

reason to test a shared mechanism and not evidence that one exists. Delirium risk was higher with glioma or malignant tumors in five of six studies, which suggests that the phenotype may differ in size between tumor types.

The biomarker evidence needs careful reading. S100B, NSE, GFAP, tau, and NfL show that injury occurred, and their kinetics differ, with S100B acting early and NfL showing a late second peak.[52,55] GFAP is complicated in glioma by release from tumor cells.[56] Serial sampling to day 10 and a panel that pairs an early astrocytic marker with a late axonal marker therefore look more informative than a single sample, as reviews of serial sampling and of neurofilament biomarkers also suggest.[20,21] Trials used these markers as surrogates for neuroprotection, and the link to clinical benefit was weak, since lower S100B with magnesium was not accompanied by a functional difference.[67] Lower marker levels do not by themselves show protection.

The two-hit model organizes these findings (Figure 3). New DWI lesions followed 18.8% to 45.7% of resections, injury markers rose in every study that followed them, and delirium affected 4.2% to 35% of patients. Preoperative state predicted the postoperative course in several designs. The model borrows biological plausibility from studies of microglial priming, in which a brain already exposed to inflammation responds more strongly to a later systemic challenge.[18,19] That literature concerns neurodegeneration and not tumors, and we do not claim it applies. What the mapped evidence cannot yet show is the interaction, meaning whether the same perioperative injury does more harm in a brain that is already vulnerable. A direct test would enroll patients with a defined tumor type and profile the four domains before surgery, sample injury and inflammatory markers serially, obtain early DWI, and follow cognition, with the interaction between preoperative profile and injury as the primary analysis. Modifiable

second-hit factors such as anesthetic technique and hemodynamic management showed favorable signals in small trials, mostly with surrogate endpoints, and should be studied inside that design.

The instrument-focused stratum matters for Indonesia in particular. No core study came from Indonesia, and none used MoCA-INA or NANO-Ina. The Indonesian studies we found were cross-sectional, with 30 to 83 participants each, and measured agreement, score distributions, and sensitivity to short-term change.[68,70-72] MoCA-INA mean scores were lower than MMSE scores in older adults[71] and in patients with brain tumors,[70] as the greater sensitivity reported for the parent MoCA would predict.[22,23] Responsiveness to craniotomy and the ability of either instrument to separate the first hit from the second remain untested, and a perioperative validation of both in Indonesian neuro-oncology cohorts is a concrete next step.

A scoping review maps evidence and does not estimate effects, so readers who need effect estimates for frailty indices should turn to the existing meta-analyses.[7] Three of the seven review questions ended partly answered, which is an informative scoping result because it directs future work. Several limitations apply. We did not appraise risk of bias, and the evidence level reflects design only, so a level 2 study may still be at high risk of bias. The direction-of-effect tallies rest on the authors' own conclusions and our coding. More than half of the core studies sampled mixed tumor types without stratified results. Screening was limited to English-language reports, and Google Scholar searches are difficult to reproduce. The instrument-focused stratum was added after charting began, and two of its studies enrolled participants without tumors. Finally, we did not search for studies of general brain frailty outside neuro-oncology.

Studies should define and test a brain-specific vulnerability profile in single-tumor cohorts, starting with glioma and meningioma. A core outcome set should replace the nine neurological definitions and the many cognitive tests, with MoCA-based screening, a domain-

specific test, and NANO as obvious candidates. Biomarker work should add cytokine measurement to injury markers and should compare meningioma with glioma. Imaging studies should test brain frailty markers and muscle measures on routine scans, and the interaction between the two hits should be the primary question of at least one prospective cohort.

Conclusions

The mapped literature does not define tumor-associated brain frailty, but it supplies the parts of a definition. Systemic frailty, cognition, neurological function, and imaging each carry evidence of vulnerability before surgery, and perioperative brain injury adds a second hit in a large share of operations. Neuroinflammation is documented in glioma and is a plausible but untested shared mechanism in meningioma. None of the 70 core studies used MoCA-INA or NANO-Ina. The evidence is observational and heterogeneous, and the interaction between the two hits is untested. Prospective cohorts that pair preoperative domain profiling with serial injury and inflammatory markers are needed to turn this working construct into a measurable and testable one.

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Figure Legends

Figure 1. PRISMA 2020 flow diagram of study identification, screening, and inclusion, with databases and the preprint server on the left and other methods on the right. Other methods are Google Scholar, citation searching, and sources supplied by the clinical team.

Figure 2. Evidence map of domains against tumor category. Bubble size is the number of core studies and shading is the share of graded studies at Oxford Centre for Evidence-Based Medicine level 2. Trial protocols are excluded from the bubbles but counted in the column totals.

Figure 3. Conceptual two-hit model. The tumor supplies the first hit and perioperative brain injury the second. No core study tested the interaction between the hits. DWI, diffusion-weighted imaging. GFAP, glial fibrillary acidic protein. NfL, neurofilament light chain.

List of Supplemental Digital Content

Supplemental digital content 1

Table showing the completed PRISMA-ScR checklist

SDC 1.xlsx

Supplemental digital content 2

Table showing the database-specific search strategies and record counts

SDC 2.xlsx

Supplemental digital content 3

Table reconciling the original screening tally with the final included studies

SDC 3.xlsx

Supplemental digital content 4

Table showing the 31-item charting form for all 76 studies

SDC 4.xlsx

Supplemental digital content 5

Table showing the coding of studies by evidence stream, design, tumor category, and domain

SDC 5.xlsx

Supplemental digital content 6

Table and figure showing the characteristics of the 70 core studies

SDC 6.docx

Supplemental digital content 7

Figure showing the four candidate domains, instruments, and gaps

SDC 7.docx

Supplemental digital content 8

Table showing inflammation, injury, and vulnerability evidence by tumor category

SDC 8.docx

Supplemental digital content 9

Table showing the biomarkers measured in the core evidence base

SDC 9.docx

Supplemental digital content 10

Table showing the evidence on perioperative brain injury as a second hit

SDC 10.docx

Supplemental digital content 11

Table showing the instrument-focused stratum

SDC 11.docx